Cost Optimization Research – Storing Models in GCS vs Persistent Vertex AI Endpoints

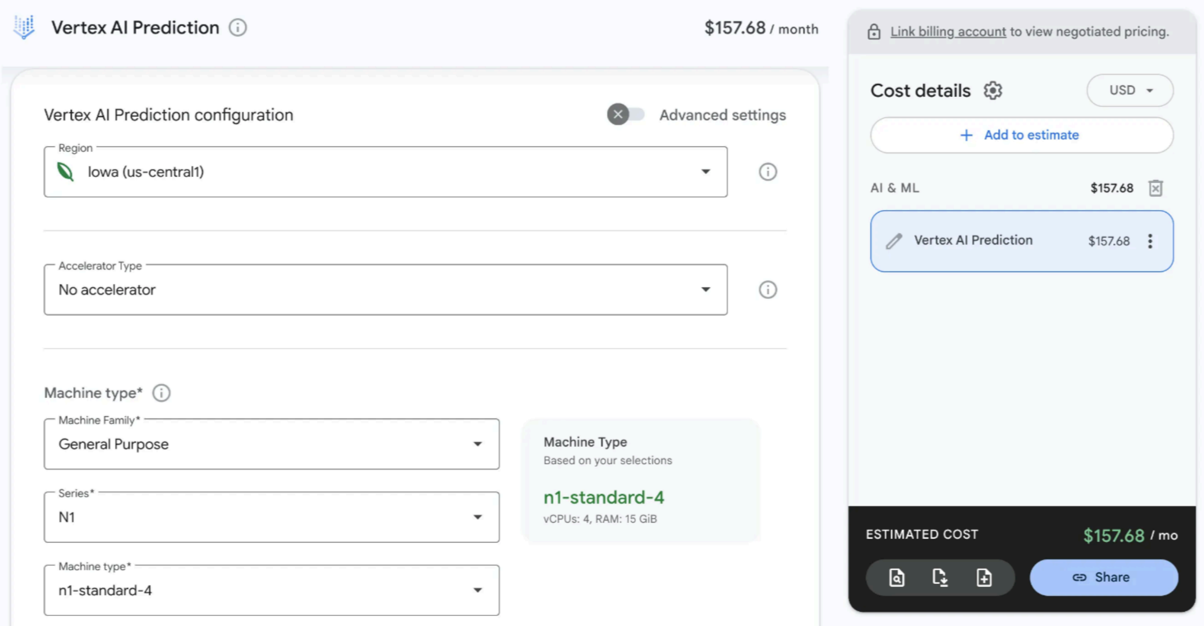
1. Abstract

Instead of paying for idle Vertex AI endpoints, store your model in GCS and load it only when needed with Cloud Run, Vertex AI jobs, or GKE — saving costs while keeping good performance through smart caching.

2. Problem Statement

Persistent Vertex AI endpoints charge continuously, even when idle. For models with low or irregular usage, it's cheaper to keep them in GCS and load them only when needed for predictions.

Cost Component	Persistent Vertex AI Endpoint	GCS + On-Demand Compute
Model Deployment Cost	$0 – Included in endpoint	$0 – Model stored in GCS (standard storage rates apply)
Compute Cost	Charged hourly, per replica (24/7), even if idle	Charged only for actual usage time (Cloud Run, Custom Job, etc.)
Storage Cost	Implicit in endpoint; no transparency	$0.026/GB/month (Standard GCS pricing)
Minimum Deployment Units	1 replica minimum	No replicas when idle
Auto-scaling	Yes, but minimum nodes still incur cost	Scales to zero when idle (Cloud Run, K8s, etc.)
Cold Start Penalty	None, always running	Present
Monthly Cost Example	~$160/month for n1-standard-4 (1 replica) scenario	<$5/month if accessed a few times and using Cloud Run



GCS Storage (e.g., 1GB model)

- **$0.026/GB/month (standard)**

North America | South America | Europe | Middle East | Asia | Africa | Australia

Location	Standard storage (per GB per Month)	Nearline storage (per GB per Month)	Coldline storage (per GB per Month)	Archive storage (per GB per Month)	Anywhere Cache storage (per GB per Hour)
Iowa (`us-central1`)	$0.020	$0.010	$0.004	$0.0012	$0.0003
South Carolina (`us-east1`)	$0.020	$0.010	$0.004	$0.0012	$0.0003
Northern Virginia (`us-east4`)	$0.023	$0.013	$0.006	$0.0025	$0.0004
Columbus (`us-east5`)	$0.020	$0.010	$0.004	$0.0012	$0.0003
Oregon (`us-west1`)	$0.020	$0.010	$0.004	$0.0012	$0.0003
Los Angeles (`us-west2`)	$0.023	$0.016	$0.007	$0.0025	
Salt Lake City (`us-west3`)	$0.023	$0.016	$0.007	$0.0025	$0.0004
Las Vegas (`us-west4`)	$0.023	$0.013	$0.006	$0.0025	$0.0004
Dallas (`us-south1`)	$0.020	$0.010	$0.004	$0.0012	$0.0003
Montréal (`northamerica-northeast1`)	$0.023	$0.013	$0.007	$0.0025	
Toronto (`northamerica-`	$0.023	$0.013	$0.007	$0.0025	

Cloud Run CPU time (assumed 20 inferences/day):

- CPU: $0.00002400 / vCPU-sec
- Assume each inference takes 3s → 20 * 3s = 60s/day

- 60s/day * 30 days = 1,800s = **$0.0432/month**

Services (Requests-based billing)

Services with request-based billing during billed instance time

Free tier (based on us-central1 active pricing):

- CPU - First **180,000 vCPU-seconds free** per month
- RAM - First **360,000 GiB-seconds free** per month
- Requests - **2 million requests free** per month

Iowa (us-central1) Show discount options

Resource	Type	Default* (USD)	Cloud Run CUD - 1 Year* (USD)	Cloud Run CUD - 3 Year* (USD)	Compute Flexible CUD - 1 Year* (USD)	Compute Flexible CUD - 3 Year* (USD)
CPU (per vCPU-second)	Active time	$0.000024	$0.00001992	$0.00001992	$0.00001992	$0.00001992
	Idle time (Min instance[1])	$0.0000025	$0.000002075	$0.000002075	$0.000002075	$0.000002075
Memory (per GiB-second)	Active time	$0.0000025	$0.000002075	$0.000002075	$0.000002075	$0.000002075
	Idle time (Min instance[1])	$0.0000025	$0.000002075	$0.000002075	$0.000002075	$0.000002075
Requests (per 1,000,000)	N/A	$0.40	$0.332	$0.332	$0.332	$0.332

GCS + CloudRun Total: Under **$1/month** for infrequent usage.

For reference pricing, see:

- https://cloud.google.com/products/calculator?hl=en

3. Load Time, Caching & Scalability

Performance Impact

Factor	Persistent Endpoint	GCS + On-Demand
Load Time	Instant (model pre-loaded)	Cold start (load from GCS) ~1–5 seconds
Caching	Internal caching handled by Vertex AI	Manual caching needed
Scalability	Built-in autoscaling	Must be architected with scalable infra (Cloud Run, GKE, etc.)
Startup Latency	Low (<100ms)	Medium-high (cold starts, GCS fetch, model load

Implementation Patterns

Pattern 1: GCS + Cloud Run

- Store model files in GCS
- Deploy a FastAPI or Flask service in Cloud Run
- On request, fetch model from GCS and run inference
- Cache model in memory

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Client → Cloud Run → Load Model from GCS → Run Inference → Response
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Pattern 2: GCS + GKE Deployment

- Load model on pod startup
- Good for larger models
- Can use Redis for caching

Pattern 3: Batch Inference with Vertex AI Custom Jobs

- Triggered via Cloud Scheduler or Pub/Sub
- Loads model from GCS and runs batch predictions
- Output written to GCS

Recommendations

Scenario	Recommended Approach	Rationale
Real-time, critical workload	Vertex AI Persistent Endpoint	Low latency, managed scalability
Infrequent predictions	GCS + Cloud Run	Cost-efficient, scalable, easy to deploy
Research / internal tools	GCS + Cloud Functions	Lightweight, serverless, very low cost
Large-scale batch predictions	GCS + Vertex AI Custom Job	Optimized for processing large datasets
GPU-based inference	GCS + GKE or Vertex AI Custom Job	GPU available; flexibility to scale workloads

Conclusion

For models that aren't used often, it's cheaper to keep them in GCS and load them only when needed with Cloud Run or Vertex AI jobs. For real-time apps, always-on endpoints are better. A mix of both can balance cost and performance.

References

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